PHYS 143 – Mechanics and Thermal Physics

Rotation of Rigid Bodies

How do we describe and predict the motion of a spinning body, and what plays the role of velocity, mass, force, work and momentum for rotation?

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By the end of this week, you should be able to:

1. Describe rotation using angular position, velocity and acceleration, and relate them to the linear motion of a point on the body.
2. Calculate moments of inertia and torques, and use $\sum \tau = I\alpha$ to find angular accelerations.
3. Use work, power and energy methods for rotating and rolling bodies.
4. Use angular momentum and its conservation, including when the moment of inertia changes.

Two contacts:

A (Tuesday): DESCRIBE, then MODEL
How do we describe the rotation of a rigid body, and what is rotational inertia?

B (Wednesday): MODEL, then APPLY
What turns a body, and how do work, energy and angular momentum describe it?

The detailed outcomes are in the checklist at the end of the lecture. **Reading:** Young and Freedman, Ch. 9, §9.1–9.6 and Ch. 10, §10.1, 10.2, 10.4–10.6.

Practice: concept check → worked example → independent problem → application problem. **Labels:** **CORE** you should be able to do this; **PRACTICE** use it to develop the skill; **REFERENCE** keep it for later use.

Where we start: a five-minute check

Not marked. Answer individually, then check the answers.

1. Find the circumference of a circle of radius $0.40\,\mathrm{m}$, and the angle in degrees of one quarter of the circle.
2. A car starts from rest and accelerates uniformly at $2.0\,\mathrm{m/s^2}$ for $5.0\,\mathrm{s}$. Find its final speed and the distance covered.
3. A $0.50\,\mathrm{kg}$ stone moves in a horizontal circle of radius $0.80\,\mathrm{m}$ at $4.0\,\mathrm{m/s}$. Find its acceleration and the net force, with their direction.
4. Vectors $\vec{\boldsymbol{A}}$ (3.0 units) and $\vec{\boldsymbol{B}}$ (4.0 units) are 30° apart. Find the magnitude of $\vec{\boldsymbol{A}} \times \vec{\boldsymbol{B}}$ and say how its direction is found.
5. A $20\,\mathrm{kg}$ crate is pushed $3.0\,\mathrm{m}$ by a $40\,\mathrm{N}$ force in $6.0\,\mathrm{s}$. Find the work done and the average power.
6. A $2.0\,\mathrm{kg}$ ball falls from rest through $5.0\,\mathrm{m}$. Use energy to find its speed on landing.

$C = 2\pi r = 2\pi(0.40) = 2.5\,\mathrm{m}$; a quarter of a circle is 90°.

$v = at = (2.0)(5.0) = 10\,\mathrm{m/s}$; $x = \frac{1}{2}at^2 = 25\,\mathrm{m}$.

$a = v^2/r = 16/0.80 = 20\,\mathrm{m/s^2}$ toward the centre; $F = ma = 10\,\mathrm{N}$ toward the centre.

$|\vec{\boldsymbol{A}} \times \vec{\boldsymbol{B}}| = AB\sin 30^\circ = 6.0$ units; the direction is perpendicular to both vectors, by the right-hand rule.

$W = Fs = (40)(3.0) = 120\,\mathrm{J}$; $P = W/t = 20\,\mathrm{W}$.

$v = \sqrt{2gh} = \sqrt{2(9.80)(5.0)} = 9.9\,\mathrm{m/s}$.

Week 1 (the cross product), Week 2 (constant acceleration), Week 3 (circular motion), Week 4 (work and power) and Week 5 (energy) are all reused this week.

Contact A: Describe, then Model

How do we describe the rotation of a rigid body, and what is rotational inertia?

Weeks 2 to 5

a particle with a position x, velocity v, acceleration a.

Where is it, how fast, how does that change?

Week 8

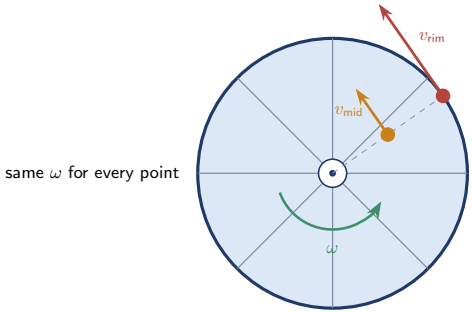
a *rigid body* (tyre, CD, fan, turbine) turning about an axis. Every point moves in a circle about the axis.

What single description covers all of its points?

Opening question

A bicycle wheel of radius $0.35\,\mathrm{m}$ turns at 120 revolutions per minute. A point on the rim and a point halfway from the axle to the rim are marked. Which point has the greater *angular* velocity? Which has the greater *speed*, and by what factor?

Two points on one wheel



Predict first

Sketch the wheel and the velocity of each point. Predict: the ratio of the two speeds, and the speed of the rim in m/s (a brisk walk is about $1.5\,\text{m/s}$).

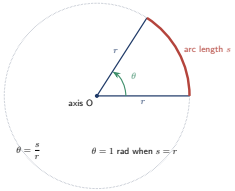
Defining the vocabulary of rotation

CORE

Quantity	What it is	What it measures	SI unit	Representation
Rigid body	an object whose parts keep fixed distances from one another	a body that turns as a whole about an axis	—	a shape with a marked axis
Angular position θ	the angle turned from a reference line	how far the body has rotated	rad	arc length over radius, s/r
Angular velocity ω	rate of change of θ	how fast, and in which sense, the body rotates	rad/s	slope of $\theta(t)$; vector along the axis
Angular acceleration α	rate of change of ω	how quickly the rotation speeds up or slows down	$\mathrm{rad/s^2}$	slope of $\omega(t)$

Interpret: ω answers *how fast, and which way, is it turning?* Every point of the body has the *same* ω and α.

Angle in radians



Rule: angle in radians

$$\theta = \frac{s}{r} \qquad 1 \text{ rev} = 2\pi \text{ rad} = 360^\circ$$

s is the arc length on a circle of radius r.

$$\begin{aligned} 1\,\text{rad} &= 57.3^\circ \\ 1\,\text{rev/s} &= 2\pi\,\text{rad/s} \\ 1\,\text{rpm} &= \frac{2\pi}{60}\,\text{rad/s} = 0.105\,\text{rad/s} \end{aligned}$$

Physical meaning

A radian is a ratio of two lengths and has no dimension; "rad" is a label for the unit used. Every formula that links an angle to a length needs **radians**.

Decide first, using the wheel on the previous slides

1. Do all points of a turning wheel have the same angular velocity? The same speed?
2. A wheel turns at constant ω. Is any point of it accelerating?
3. Can ω be zero at an instant while α is not?
4. Why do we measure angles in radians?

Answers

1. **Same ω; speed grows with the distance from the axis.**
2. **Yes.** Every point moves in a circle, so it has $a_{\text{rad}} = \omega^2 r$ toward the axis.
3. **Yes.** A flywheel that is momentarily at rest while reversing.
4. **A radian is a ratio of lengths,** s/r**,** so $s = r\theta$ and $v = r\omega$ hold without extra factors.

Rule: angular velocity and angular acceleration

$$\Delta\theta = \theta_2 - \theta_1 \qquad \omega_{\text{av}} = \frac{\Delta\theta}{\Delta t} \qquad \omega = \frac{\mathrm{d}\theta}{\mathrm{d}t}$$

$$\alpha_{\text{av}} = \frac{\Delta\omega}{\Delta t} \qquad \alpha = \frac{\mathrm{d}\omega}{\mathrm{d}t} = \frac{\mathrm{d}^2\theta}{\mathrm{d}t^2}$$

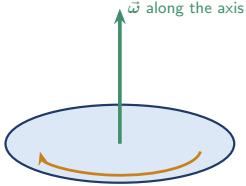
Units: ω in rad/s, α in rad/s^2. For rotation about a fixed axis choose a positive sense (anticlockwise is positive): ω and α are then signed numbers.

Physical meaning

ω is the slope of $\theta(t)$. α is the slope of $\omega(t)$. This is Week 2 with $x \to \theta$, $v \to \omega$, $a \to \alpha$.

Common error

$\omega > 0$ and $\alpha < 0$ together mean the body is *slowing down* while still turning in the positive sense, not that it has stopped.



Right-hand rule: curl the fingers in the sense of rotation, the thumb gives the direction of $\vec{\omega}$.

Speeding up: $\vec{\alpha}$ along $\vec{\omega}$. Slowing down: $\vec{\alpha}$ opposite to $\vec{\omega}$.

rotation: anticlockwise seen from above

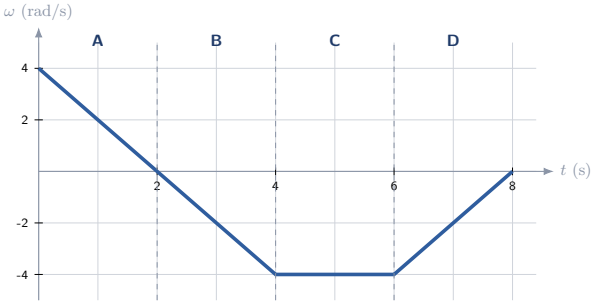
Rule: the direction of $\vec{\omega}$ and $\vec{\alpha}$

Curl the fingers of the right hand the way the body turns: the thumb points along $\vec{\omega}$ (the corkscrew rule). $\vec{\omega}$ and $\vec{\alpha}$ both lie along the fixed axis.

For a fixed axis a sign is enough: ω is positive for anticlockwise rotation as seen from the tip of the axis vector, and negative for clockwise.

Concept check: reading an $\omega(t)$ graph

CORE



A wheel's angular velocity is plotted in four segments, A to D (anticlockwise is positive).

Answer for each segment, with a reason

(a) The sense of rotation. (b) The sign of α. (c) Is the wheel speeding up or slowing down? (d) The angle turned in segment A.

Reading an $\omega(t)$ graph: answers

Segment	Sense (ω)	α (slope)	Speed	$\Delta\theta$ (area)
A (0–2 s)	anticlockwise	$-2.0\,\mathrm{rad/s^2}$	slowing down	$+4.0\,\mathrm{rad}$
B (2–4 s)	clockwise	$-2.0\,\mathrm{rad/s^2}$	speeding up	$-4.0\,\mathrm{rad}$
C (4–6 s)	clockwise	0	constant	$-8.0\,\mathrm{rad}$
D (6–8 s)	clockwise	$+2.0\,\mathrm{rad/s^2}$	slowing down	$-4.0\,\mathrm{rad}$

Rule of thumb

ω and α with the *same* sign: speeding up. *Opposite* signs: slowing down. The angle turned is the area under the ω–t graph; the net angle over $8\,\mathrm{s}$ is $-12\,\mathrm{rad}$ (clockwise).

Question

A turbine disc turns through an angle $\theta(t) = 3.0t^2 - 0.20t^3$ (θ in radians, t in seconds) while it is being shut down. Find (a) θ, ω and α at $t = 2.0\,\mathrm{s}$; (b) the time at which ω is greatest, and its value; (c) the average angular velocity between $t = 2.0\,\mathrm{s}$ and $t = 4.0\,\mathrm{s}$.

Predict first

Sketch $\omega(t)$ before calculating. Does ω increase all the time? What is α at the instant ω is greatest? Will ω_{av} for the interval be the mean of the two end values?

Set up: differentiate: $\omega = \dfrac{\mathrm{d}\theta}{\mathrm{d}t} = 6.0t - 0.60t^2$ and $\alpha = \dfrac{\mathrm{d}\omega}{\mathrm{d}t} = 6.0 - 1.2t$.

(a) At $t = 2.0\,\mathrm{s}$: $\theta = 3.0(4.0) - 0.20(8.0) = 10.4\,\mathrm{rad}$; $\omega = 12.0 - 2.4 = 9.6\,\mathrm{rad/s}$; $\alpha = 6.0 - 2.4 = 3.6\,\mathrm{rad/s^2}$.

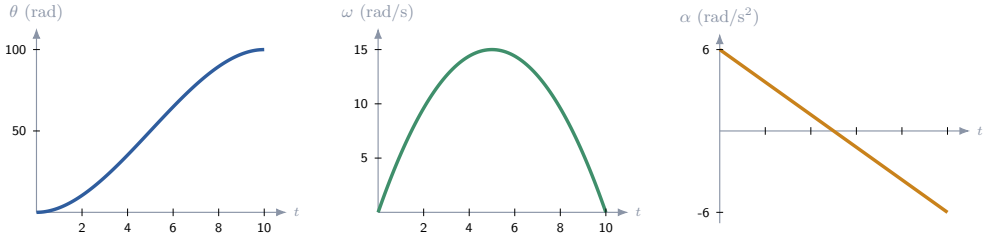
(b) ω is greatest where $\alpha = 0$: $6.0 - 1.2t = 0 \Rightarrow t = 5.0\,\mathrm{s}$, and $\omega_{\max} = 30 - 15 = 15\,\mathrm{rad/s}$.

(c) $\theta(4.0) = 48.0 - 12.8 = 35.2\,\mathrm{rad}$, so $\omega_{\mathrm{av}} = \dfrac{35.2 - 10.4}{2.0} = 12.4\,\mathrm{rad/s}$.

Evaluate: $\omega(2.0) = 9.6$ and $\omega(4.0) = 14.4\,\mathrm{rad/s}$; their mean is $12.0\,\mathrm{rad/s}$, not 12.4: the average is not the mean because α is not constant. The disc stops at $t = 10\,\mathrm{s}$ ($\omega = 0$). ✓

The three graphs of $\theta(t)$

CORE



ω is the slope of the left graph; α is the slope of the middle graph. ω peaks where α crosses zero ($t = 5.0\,\mathrm{s}$), and $\omega = 0$ where θ stops increasing ($t = 10\,\mathrm{s}$). Between 0 and $5\,\mathrm{s}$, ω and α have the same sign (speeding up); after $5\,\mathrm{s}$ the signs differ (slowing down).

Rule: constant angular acceleration

$$\omega = \omega_0 + \alpha t \qquad \theta - \theta_0 = \omega_0 t + \tfrac{1}{2}\alpha t^2$$
$$\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0) \qquad \theta - \theta_0 = \tfrac{1}{2}(\omega_0 + \omega)t$$

valid **only** when α is constant, with θ in radians.

Week 2 (linear)	Week 8 (rotation)	Variable missing
$v = v_0 + at$	$\omega = \omega_0 + \alpha t$	θ
$x - x_0 = v_0 t + \frac{1}{2}at^2$	$\theta - \theta_0 = \omega_0 t + \frac{1}{2}\alpha t^2$	ω
$v^2 = v_0^2 + 2a(x - x_0)$	$\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0)$	t
$x - x_0 = \frac{1}{2}(v_0 + v)t$	$\theta - \theta_0 = \frac{1}{2}(\omega_0 + \omega)t$	α

Question

A washing-machine drum spinning at 900 rpm is braked uniformly to rest in $12\,\mathrm{s}$. Find (a) its angular acceleration, and (b) the number of revolutions it makes while stopping.

Predict first

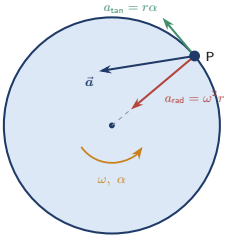
Sketch $\omega(t)$. Will α be positive or negative? At 900 rpm $= 15$ rev/s for $12\,\mathrm{s}$ the drum would make 180 revolutions at constant speed: will the actual number be more or less than that?

Set up: anticlockwise positive; $\omega_0 = 900 \times \dfrac{2\pi}{60} = 94.2\,\text{rad/s}$, $\omega = 0$, $t = 12\,\text{s}$; α constant.

(a) $\omega = \omega_0 + \alpha t \Rightarrow \alpha = \dfrac{0 - 94.2}{12} = -7.85\,\text{rad/s}^2$.

(b) The equation without α: $\theta - \theta_0 = \frac{1}{2}(\omega_0 + \omega)t = \frac{1}{2}(94.2)(12) = 565\,\text{rad}$, which is $\dfrac{565}{2\pi} = 90.0$ revolutions.

Evaluate: $\alpha < 0$ as predicted (slowing in the positive sense). The drum makes 90 revolutions, half of 180, because the average angular velocity of a uniform slowing is half the initial value. Check with $\theta - \theta_0 = \omega_0 t + \frac{1}{2}\alpha t^2 = 1131 - 565 = 565\,\text{rad}$. ✓



Rule: a point at distance r from the axis

$$s = r\theta \quad v = r\omega$$

$$a_{\text{tan}} = r\alpha \quad a_{\text{rad}} = \omega^2 r = \frac{v^2}{r}$$

with θ, ω, α in **radians**.

Physical meaning

Every point has the same ω and α; the speed and the accelerations grow in proportion to r. a_{tan} changes the speed; a_{rad} (Week 3, centripetal acceleration) changes the direction. The magnitude of the total acceleration is $a = \sqrt{a_{\text{tan}}^2 + a_{\text{rad}}^2}$.

Question

A wheel of radius $0.45\,\mathrm{m}$ starts from rest and speeds up with a constant angular acceleration of $2.4\,\mathrm{rad/s^2}$. At $t = 3.0\,\mathrm{s}$ find, for a point on the rim, (a) the angular velocity, (b) the speed, (c) the tangential and radial accelerations, and (d) the magnitude of the total acceleration and its angle with the radius.

Predict first

Sketch the point and its two acceleration components. At that instant, which component is the larger, and is the total close to one of them?

A point on a speeding-up wheel: solution

Set up: $\alpha = 2.4\,\text{rad/s}^2$ (constant), $\omega_0 = 0$, $r = 0.45\,\text{m}$, $t = 3.0\,\text{s}$.

(a) $\omega = \alpha t = 7.2\,\text{rad/s}$. **(b)** $v = r\omega = (0.45)(7.2) = 3.24\,\text{m/s}$.

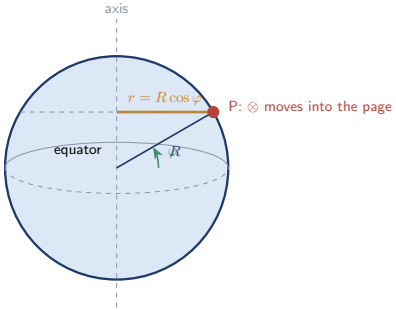
(c) $a_{\text{tan}} = r\alpha = (0.45)(2.4) = 1.08\,\text{m/s}^2$; $a_{\text{rad}} = \omega^2 r = (7.2)^2(0.45) = 23.3\,\text{m/s}^2$.

(d) $a = \sqrt{1.08^2 + 23.3^2} = 23.4\,\text{m/s}^2$; the angle with the inward radius is $\arctan\dfrac{1.08}{23.3} = 2.7^\circ$.

Evaluate: $a_{\text{rad}} \gg a_{\text{tan}}$ at this instant, so the total points almost at the centre, as predicted.
Check: $a_{\text{rad}} = v^2/r = 3.24^2/0.45 = 23.3\,\text{m/s}^2$. ✓

Independent problem

The Earth turns once in $24.0\,\mathrm{h}$ (a simplification: the true period is $23.93\,\mathrm{h}$). Accra is at latitude $5.6°$ N. The Earth's radius is $6.37 \times 10^6\,\mathrm{m}$. Find (a) the Earth's angular velocity, (b) the speed of a person in Accra because of the rotation, and (c) the person's centripetal acceleration as a fraction of g.



Predict first

Sketch the circle of latitude: what is its radius for a point near the equator? Is the speed closer to $4.6\,\mathrm{m/s}$, $46\,\mathrm{m/s}$ or $460\,\mathrm{m/s}$?

Your turn: solution

Set up: the point moves on a circle of latitude of radius $r = R\cos\varphi = (6.37\times10^6)\cos 5.6^\circ = 6.34\times10^6\,\text{m}$.

(a) $\omega = \dfrac{2\pi}{T} = \dfrac{2\pi}{(24.0)(3600)} = 7.27\times10^{-5}\,\text{rad/s}$.

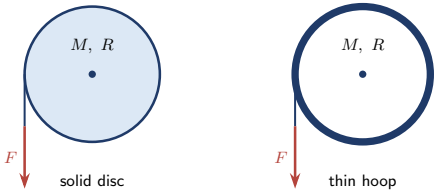
(b) $v = r\omega = (6.34\times10^6)(7.27\times10^{-5}) = 461\,\text{m/s}$.

(c) $a_{\text{rad}} = \omega^2 r = (7.27\times10^{-5})^2(6.34\times10^6) = 0.0335\,\text{m/s}^2$, which is $\dfrac{0.0335}{9.80} = 3.4\times10^{-3}\,g$.

Evaluate: the answer is of order $460\,\text{m/s}$, faster than sound in air, yet we feel nothing because the air and the ground move with us at constant velocity, and a_{rad} is only about 0.3% of g. ✓

Common mistakes and the correct habit

Mistake	Instead
Using degrees or rev/min in $v = r\omega$ or $s = r\theta$.	Convert to radians and rad/s first.
Measuring r from the centre of the body.	r is the distance from the axis.
Forgetting a_{rad} for a point on a wheel at constant ω.	$a_{\text{rad}} = \omega^2 r$ is still present.
Using the constant-α equations when α varies.	Differentiate or integrate instead.
Reading $\omega > 0$, $\alpha < 0$ as stopped.	It means slowing while still turning.
Taking I about the wrong axis, or using MR^2 for every body.	Name the axis; use the table or the parallel-axis theorem.



Poll

A cord is wound on the rim of a solid disc and of a thin hoop. Both have the same mass M and radius R, both are free to turn about their centres, and the cord is pulled with the same force F. Which one reaches the larger angular acceleration, or are they equal?

Predict first

Same force, same lever arm. What property of the body, other than its mass, must matter? Sketch where the mass of each body sits.

Defining the vocabulary of rotational dynamics

Quantity	What it is	What it measures	SI unit	Representation
Moment of inertia I	the sum $\sum m_i r_i^2$ about a chosen axis	how strongly the body resists a change of rotation	$\mathrm{kg\,m^2}$	the way the mass is spread about the axis
Torque τ	the turning effect of a force about an axis	how effectively a force turns the body	N m	force $\times$ lever arm; a vector along the axis
Rotational kinetic energy	the energy of the turning motion	the work needed to set the body turning	J	$\frac{1}{2} I \omega^2$

Interpret: I plays the role of mass and torque the role of force: the question is *which axis, and how is the mass spread about it?*

Treat the body as many small particles. Particle i has mass m_i at distance r_i from the axis, so its speed is $v_i = r_i\omega$:

$$K = \sum_i \tfrac{1}{2} m_i v_i^2 = \sum_i \tfrac{1}{2} m_i (r_i\omega)^2 = \tfrac{1}{2}\Big(\sum_i m_i r_i^2\Big)\omega^2.$$

Rule: kinetic energy of rotation and moment of inertia

$$K = \tfrac{1}{2} I\omega^2 \qquad I = \sum_i m_i r_i^2 \qquad [I] = \mathrm{kg\,m^2}$$

r_i is the distance from the **axis of rotation**, so I depends on the axis chosen.

Physical meaning

I plays the role of mass for rotation: it measures how hard it is to change the rotation. Mass far from the axis counts more, in proportion to r^2.

Common error

I is not a property of the body alone. The same body has different values of I about different axes.

Centre of mass of particles on a line

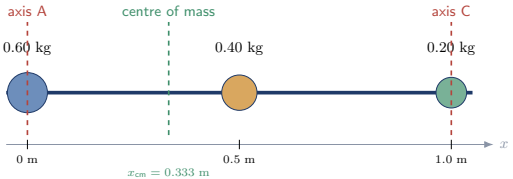
$$x_{\text{cm}} = \frac{m_1 x_1 + m_2 x_2 + \cdots}{m_1 + m_2 + \cdots} = \frac{\sum m_i x_i}{M}$$

The weighted average position of the mass: the point where the body balances. For a uniform body of symmetric shape it lies at the centre of symmetry.

We need only this definition today, to compare axes; the supplementary notes treat the centre of mass and its motion. Two particles of $2.0\,\text{kg}$ at $x = 0$ and $6.0\,\text{kg}$ at $x = 4.0\,\text{m}$ give $x_{\text{cm}} = \dfrac{0 + (6.0)(4.0)}{8.0} = 3.0\,\text{m}$: it lies nearer the heavier particle.

Question

Three small masses are fixed on a light rigid rod (negligible mass) as shown. Find I for an axis perpendicular to the rod through (a) mass A, (b) mass C, and (c) the centre of mass. Then find K for each axis when $\omega = 6.0\,\mathrm{rad/s}$.



Predict first

Sketch the three axes. About which axis do you expect the smallest I, and about which the largest? Is I about C close to I about A?

One body, three axes: solution

Set up: $M = 1.20\,\text{kg}$; $x_{\text{cm}} = \dfrac{(0.60)(0) + (0.40)(0.50) + (0.20)(1.00)}{1.20} = 0.333\,\text{m}$.
$I = \sum m_i r_i^2$ with r_i measured from each axis.

(a) Axis A: $I_A = 0 + (0.40)(0.50)^2 + (0.20)(1.00)^2 = 0.30\,\text{kg}\,\text{m}^2$.
(b) Axis C: $I_C = (0.60)(1.00)^2 + (0.40)(0.50)^2 + 0 = 0.70\,\text{kg}\,\text{m}^2$.
(c) Centre of mass: $I_{\text{cm}} = (0.60)(0.333)^2 + (0.40)(0.167)^2 + (0.20)(0.667)^2 = 0.167\,\text{kg}\,\text{m}^2$.

Kinetic energy $K = \frac{1}{2} I\omega^2$ at $6.0\,\text{rad/s}$: $5.4\,\text{J}$ (A), $12.6\,\text{J}$ (C), $3.0\,\text{J}$ (centre of mass).

Evaluate: I is smallest about the centre of mass and largest about the far end, as predicted.
Pattern: $I_A - I_{\text{cm}} = 0.133 = M(0.333)^2$ and $I_C - I_{\text{cm}} = 0.533 = M(0.667)^2$.

Rule: parallel-axis theorem

$$I = I_{\text{cm}} + Md^2$$

I_{cm} is the moment of inertia about the axis through the centre of mass; d is the distance between that axis and the parallel axis; M is the total mass.

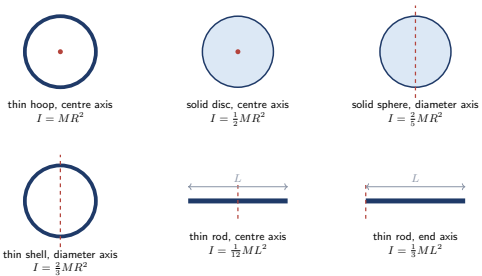
Check on the example:

$I_A = 0.167 + (1.20)(0.333)^2 = 0.300$
$I_C = 0.167 + (1.20)(0.667)^2 = 0.700$ ✓

Physical meaning

For a given direction of the axis, I is smallest about the axis through the centre of mass. Moving the axis away can only increase I. The proof is in the supplementary notes.

Moments of inertia of standard bodies



For the same M and R: hoop $>$ disc, and shell $>$ solid sphere. The more mass sits far from the axis, the larger the I.

rectangular plate, axis through the centre, perpendicular to the plate (sides a, b)	$\frac{1}{12}M(a^2+b^2)$	thin plate, axis along an edge (the other side a)	$\frac{1}{3}Ma^2$
hollow cylinder (radii R_1, R_2), axis along its centre	$\frac{1}{2}M(R_1^2+R_2^2)$	solid cylinder; thin-walled hollow cylinder	$\frac{1}{2}MR^2$; MR^2

The integrals that give these results are an unassessed extension in the supplementary notes.

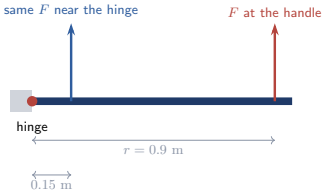
Question

A uniform thin rod of mass $0.50\,\mathrm{kg}$ and length $0.60\,\mathrm{m}$ carries a small sphere of mass $0.40\,\mathrm{kg}$ at each end (treat each sphere as a point mass). Find I for an axis perpendicular to the rod through (a) its centre and (b) one end.

Predict first

Sketch the rod with the two axes. By what factor, roughly, will I about the end exceed I about the centre: 1, 2 or 4?

(a) Centre: rod $\frac{1}{12}ML^2 = \frac{1}{12}(0.50)(0.60)^2 = 0.015$; spheres $2(0.40)(0.30)^2 = 0.072$; $I = 0.087\,\mathrm{kg\,m^2}$.
(b) End: rod $\frac{1}{3}ML^2 = 0.060$; sphere on the axis 0; far sphere $(0.40)(0.60)^2 = 0.144$; $I = 0.204\,\mathrm{kg\,m^2}$.
Evaluate: ratio 2.3, close to 2 as predicted. Check by the theorem: $0.087 + (1.30)(0.30)^2 = 0.204$. ✓



The same force F is applied perpendicular to the door, once at the handle ($0.90\,\mathrm{m}$ from the hinge) and once $0.15\,\mathrm{m}$ from the hinge.

Which opens the door faster? What if $\vec{\boldsymbol{F}}$ pointed along the door, toward the hinge?

Predict first

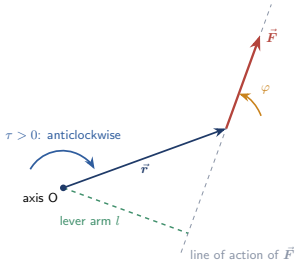
Rank the **turning effects** of these cases before you read on: the handle push, the push near the hinge, the push along the door.

Decide first

1. Can a force that acts on a body produce no torque?
2. Can the net force on a body be zero while the net torque is not?
3. Does a force of the same size always give the same torque about a given axis?

Answers

1. **Yes.** If its line of action passes through the axis.
2. **Yes.** Equal and opposite forces on opposite sides of the axis (a couple) turn the body without moving its centre.
3. **No.** Torque depends on where and at what angle the force acts: $\tau = rF\sin\varphi$.



Rule: torque about an axis O

$$\tau = rF\sin\varphi = F_\perp r = Fl$$

$$\vec{\boldsymbol{\tau}} = \vec{\boldsymbol{r}} \times \vec{\boldsymbol{F}}$$

φ is the angle between $\vec{\boldsymbol{r}}$ and $\vec{\boldsymbol{F}}$; $l = r\sin\varphi$ is the lever arm. Unit: N m.

Physical meaning

Torque is the turning effect of a force about a chosen axis: it grows with the force and with the perpendicular distance from the axis to the *line of action*. Sign: $\tau > 0$ for anticlockwise, $\tau < 0$ for clockwise. It is zero if the line of action passes through the axis ($\varphi = 0^\circ$ or 180°).

Common error

Torque is in N m and work is in joules (J). The units have the same dimensions, but the quantities differ: write torque as N m and energy as J. The direction of $\vec{\tau}$ is along the axis (right-hand rule); energy has no direction.

Net torque

For several forces about the same axis, add the *signed* torques:

$$\sum \tau = \tau_1 + \tau_2 + \cdots$$

A zero net *force* does not imply a zero net *torque*.

The door: handle push $\tau = (0.90)F$; near the hinge $\tau = (0.15)F$, six times smaller; along the door, $\varphi = 0$ and $\tau = 0$.

Contact A: what you can now do

- Convert between degrees, radians, revolutions and rpm; find ω and α from $\theta(t)$ and read the graphs.
- Use the constant-α equations, and relate a point on the body to the body: $v = r\omega$, $a_{\text{tan}} = r\alpha$, $a_{\text{rad}} = \omega^2 r$.
- Calculate $I = \sum mr^2$ (parallel axis, standard bodies) and $K = \frac{1}{2}I\omega^2$, and a torque $\tau = rF\sin\varphi$ with its sign.

Predict first

Before Wednesday, predict: a cord is pulled with the same force on the rim of a solid disc and of a thin hoop of the same M and R (the opening question of this contact). Which has the larger angular acceleration? A rod hinged at one end is released from the horizontal: does its free end accelerate faster or slower than g?

But what if... a net torque acts on a body that is free to turn? → **Newton's second law for rotation, and then work, energy and angular momentum**.

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Contact B: Model, then Apply

What turns a body, and how do work, energy and angular momentum describe a rotating body?

Predict first

1. A wheel turns at 300 rpm. Find ω in rad/s. $\omega = 300 \times \dfrac{2\pi}{60} = 31.4\,\text{rad/s}$.
2. A point on that wheel is $0.20\,\text{m}$ from the axis. Find its speed and radial acceleration. $v = r\omega = (0.20)(31.4) = 6.28\,\text{m/s}$; $a_{\text{rad}} = \omega^2 r = (31.4)^2(0.20) = 197\,\text{m/s}^2$.
3. Two $1.0\,\text{kg}$ point masses are $0.50\,\text{m}$ and $1.0\,\text{m}$ from an axis. Find I, and the torque of a $20\,\text{N}$ force perpendicular to a lever arm of $0.90\,\text{m}$.
   $I = \sum mr^2 = (1.0)(0.50)^2 + (1.0)(1.0)^2 = 1.25\,\text{kg m}^2$;
   $\tau = Fl = (20)(0.90) = 18\,\text{N m}$.

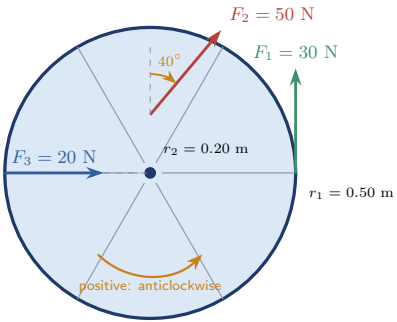
A wrench turns a bolt. Forces (a) to (d) each have magnitude $20\,\mathrm{N}$. Rank their torques about the bolt.

(a) Perpendicular to the handle, at $0.30\,\mathrm{m}$ from the bolt.
(b) At 30° to the handle, at $0.30\,\mathrm{m}$.
(c) Perpendicular to the handle, at $0.15\,\mathrm{m}$.
(d) Along the line of the handle, pointing at the bolt.
(e) Two forces of $10\,\mathrm{N}$ in opposite directions, each perpendicular to the ends of a $0.40\,\mathrm{m}$ diameter, applied to a wheel: what are the net force and the net torque?

(a) $20(0.30) = 6.0\,\mathrm{N\,m}$; (b) $20(0.30)\sin 30^\circ = 3.0\,\mathrm{N\,m}$; (c) $20(0.15) = 3.0\,\mathrm{N\,m}$; (d) 0.
Ranking: (a) $>$ (b) $=$ (c) $>$ (d). (e) The net force is zero, but the net torque is $2(10)(0.20) = 4.0\,\mathrm{N\,m}$: a **couple** turns the wheel without accelerating its centre.

Example 6: a wheel with three forces

CORE



Question

Three forces act on a wheel that turns about its axle through O. Find the torque of each about O and the net torque, and say which way the wheel tends to turn.

Predict first

Sketch the lever arm of each force. Which forces turn the wheel anticlockwise, which clockwise, and which have zero torque?

The wheel with three forces: solution

F_1: tangent to the rim, so $\varphi = 90^\circ$: $\tau_1 = +r_1 F_1 = +(0.50)(30) = +15.0\,\mathrm{N\,m}$ (anticlockwise).

F_2: $\varphi = 40^\circ$ between $\vec{r}_2$ and $\vec{F}_2$, and it turns the wheel clockwise:
$\tau_2 = -r_2 F_2 \sin 40^\circ = -(0.20)(50)(0.643) = -6.43\,\mathrm{N\,m}$.

F_3: its line of action passes through O, so $\tau_3 = 0$.

Net torque: $\sum \tau = +15.0 - 6.43 + 0 = +8.6\,\mathrm{N\,m}$: the wheel tends to turn **anticlockwise**.

Evaluate: the signs agree with the sketch. The axle supplies a force that has no torque about O, so it does not appear in $\sum \tau$ even though $\sum \vec{F} \neq 0$. ✓

Consider particle i of a rigid body turning about a fixed axis. The tangential force on it gives $F_{\text{tan},i} = m_i a_{\text{tan},i} = m_i r_i \alpha$, and its torque is

$$\tau_i = r_i F_{\text{tan},i} = m_i r_i^2 \alpha.$$

Add all the particles. The internal forces between particles cancel in pairs, and so do their torques, leaving only the external torques:

Rule: Newton's second law for rotation

$$\sum \tau_{\text{ext}} = I\alpha$$

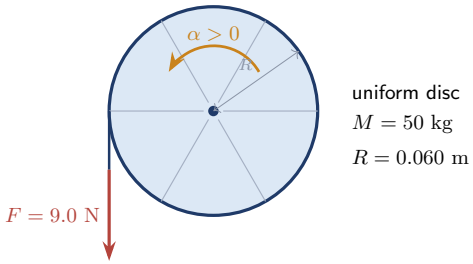
about a fixed axis, with τ and I taken about the **same** axis. α is in rad/s^2.

Physical meaning

Compare $\sum F = ma$: torque plays the role of force and I that of mass. For a given torque, a larger I gives a smaller α.

Question

A solid cylinder ($M = 50\,\mathrm{kg}$, diameter $0.120\,\mathrm{m}$) turns freely about its axle. A light cable wound round it is pulled with a constant tangential force of $9.0\,\mathrm{N}$. Starting from rest, find (a) α and the acceleration of the cable, (b) ω and the angle turned after $2.0\,\mathrm{s}$. (c) What would α be for a thin hoop of the same M and R?



Predict first

Sketch the torque and the sign of α. Will the hoop's α be larger, smaller or the same as the disc's?

Set up: anticlockwise positive; $R = 0.060\,\mathrm{m}$. $I = \frac{1}{2}MR^2 = \frac{1}{2}(50)(0.060)^2 = 0.090\,\mathrm{kg\,m^2}$. $\tau = FR = (9.0)(0.060) = 0.54\,\mathrm{N\,m}$.

(a) $\alpha = \dfrac{\tau}{I} = \dfrac{0.54}{0.090} = 6.0\,\mathrm{rad/s^2}$. The cable moves with the rim, so $a = R\alpha = 0.36\,\mathrm{m/s^2}$.

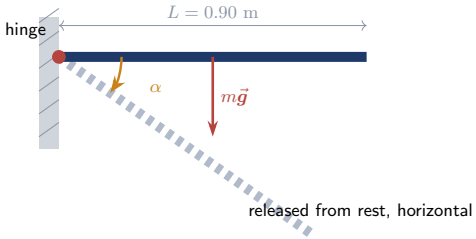
(b) $\omega = \alpha t = 12\,\mathrm{rad/s}$; $\theta = \frac{1}{2}\alpha t^2 = \frac{1}{2}(6.0)(2.0)^2 = 12\,\mathrm{rad}$ (1.9 revolutions).

(c) Hoop: $I = MR^2 = 0.18\,\mathrm{kg\,m^2}$, so $\alpha = 0.54/0.18 = 3.0\,\mathrm{rad/s^2}$, **half** that of the cylinder.

Evaluate: the same torque gives the hoop the smaller α because its mass is all at the rim, answering the opening question. $K = \frac{1}{2}I\omega^2 = \frac{1}{2}(0.090)(12)^2 = 6.5\,\mathrm{J}$, and $\tau\theta = (0.54)(12) = 6.5\,\mathrm{J}$: preview of the next contact. ✓

Independent problem

A uniform rod ($M = 1.2\,\mathrm{kg}$, $L = 0.90\,\mathrm{m}$) is hinged at one end and released from rest in a horizontal position. At the instant of release find (a) its angular acceleration, and (b) the linear acceleration of the free end.



Predict first

Which torque acts about the hinge, and where does the weight act? Is the acceleration of the free end larger or smaller than g?

Your turn: solution

Set up: axis at the hinge; clockwise positive. The hinge force has no torque about the hinge, so only the weight (at the centre) contributes. $I = \frac{1}{3}ML^2 = \frac{1}{3}(1.2)(0.90)^2 = 0.324\,\mathrm{kg\,m^2}$.

(a) $\tau = Mg\dfrac{L}{2} = (1.2)(9.80)(0.45) = 5.29\,\mathrm{N\,m}$, so $\alpha = \dfrac{\tau}{I} = \dfrac{5.29}{0.324} = 16.3\,\mathrm{rad/s^2}$.

(b) $a_{\text{tip}} = r\alpha = L\alpha = (0.90)(16.3) = 14.7\,\mathrm{m/s^2} = 1.5\,g$.

Evaluate: the free end accelerates *faster* than free fall: it is pulled down by the rest of the rod. Check: the point at $\frac{2}{3}L = 0.60\,\mathrm{m}$ has $a = (0.60)(16.3) = 9.8\,\mathrm{m/s^2} = g$. This holds only at the instant of release: as the rod turns the lever arm shrinks. ✓

Weeks 4 and 5

$W_{\text{net}} = \Delta K$ and
$K_1 + U_1 + W_{\text{other}} = K_2 + U_2$ for a particle.

Do they still work for a spinning body?

Week 8

a constant torque drives a flywheel for several seconds.

How much work is done, and how fast is energy delivered?

Opening question

A motor applies a constant torque to a flywheel that starts from rest. Is the power delivered by the motor constant in time, increasing, or decreasing? Is the work done in the second half of the time equal to, more than, or less than the work done in the first half?

Defining work, power and angular momentum

Quantity	What it is	What it measures	SI unit	Representation
Work of a torque	the energy a torque transfers as the body turns	how much a torque changes the rotational energy	J	$\tau\,\Delta\theta$
Power	the rate at which a torque delivers energy	how fast energy is supplied to the rotation	W	$\tau\omega$
Rolling without slipping	a wheel's contact point is momentarily at rest	the link between rolling and sliding speeds	m/s	$v_{\text{cm}} = R\omega$
Angular momentum L	$I\omega$, or $\vec{\boldsymbol{r}} \times \vec{\boldsymbol{p}}$ for a particle	the quantity of rotational motion about a point	$\mathrm{kg\,m^2/s}$	a vector along the axis

Interpret: the linear ideas return with $F \to \tau$, $x \to \theta$, $v \to \omega$, $m \to I$, $p \to L$.

A tangential force F at radius r moves its point of application through the arc $s = r\theta$, so the work is $W = Fs = (Fr)\theta = \tau\theta$.

Rule: work, power and the work–energy theorem for rotation

$$W = \int \tau \, \mathrm{d}\theta = \tau \, \Delta\theta \ \text{(constant } \tau\text{)} \qquad P = \frac{\mathrm{d}W}{\mathrm{d}t} = \tau\omega$$

$$W_{\text{net}} = \Delta K_{\text{rot}} = \tfrac{1}{2} I\omega_2^2 - \tfrac{1}{2} I\omega_1^2$$

for rotation about a fixed axis, with $\Delta\theta$ in radians.

	motion	inertia	cause	law	work, power	K
linear	$x,\ v,\ a$	m	F	$\sum F = ma$	$Fx,\ Fv$	$\frac{1}{2}mv^2$
rotation	$\theta,\ \omega,\ \alpha$	I	τ	$\sum \tau = I\alpha$	$\tau\theta,\ \tau\omega$	$\frac{1}{2}I\omega^2$

Question

A grindstone with $I = 2.0\,\mathrm{kg\,m^2}$ is at rest. A motor applies a constant torque of $10\,\mathrm{N\,m}$ for $8.0\,\mathrm{s}$. Find (a) α, ω and the angle turned, (b) the work done by the torque, checked against ΔK, and (c) the average power and the instantaneous power at $t = 8.0\,\mathrm{s}$.

Predict first

Sketch $P(t) = \tau\omega$ against t. Will the instantaneous power at $8.0\,\mathrm{s}$ be less than, equal to, or greater than the average power? By what factor?

A grindstone under a constant torque: solution

(a) $\alpha = \dfrac{\tau}{I} = \dfrac{10}{2.0} = 5.0\,\text{rad/s}^2$; $\omega = \alpha t = 40\,\text{rad/s}$; $\theta = \frac{1}{2}\alpha t^2 = \frac{1}{2}(5.0)(8.0)^2 = 160\,\text{rad}$ (25.5 revolutions).

(b) $W = \tau\Delta\theta = (10)(160) = 1600\,\text{J}$. Check: $\Delta K = \frac{1}{2}(2.0)(40)^2 = 1600\,\text{J}$. ✓

(c) $P_{\text{av}} = \dfrac{W}{t} = \dfrac{1600}{8.0} = 200\,\text{W}$; $P(8.0\,\text{s}) = \tau\omega = (10)(40) = 400\,\text{W}$.

Evaluate: ω grows linearly, so $P = \tau\omega$ grows linearly: the final power is exactly twice the average, and more work is done in the second half than in the first (three times as much). Same pattern as Week 4 for a constant force on a body starting from rest. ✓

Both the work–energy theorem and the Week 5 energy equation hold, with the kinetic energy of the whole body:

Rule: energy equation with rotation

$$K_1 + U_1 + W_{\text{other}} = K_2 + U_2 \qquad K = \tfrac{1}{2} M v_{\text{cm}}^2 + \tfrac{1}{2} I_{\text{cm}} \omega^2$$

For **rolling without slipping**: $v_{\text{cm}} = R\omega$.

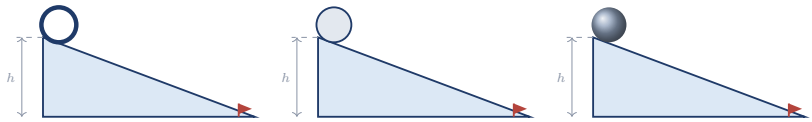
Physical meaning

A rolling body has two kinetic energies: translation of the centre of mass and rotation about it. The two are linked by $v_{\text{cm}} = R\omega$, so the rotation takes a fixed share of the energy.

Why friction does no work

The point of contact is momentarily at rest, so *static* friction does no work, and mechanical energy is conserved. If the body slips, kinetic friction acts and takes energy.

Concept check: a race down an incline



released together from rest at $h = 0.80$ m; same M and R; rolling without slipping

Poll

A thin hoop, a solid disc and a solid sphere all have the same mass and radius. Released together from rest at the same height, they roll without slipping down an incline. Which reaches the bottom first, and does the mass or the radius matter?

Predict first

Which body has the largest share of its energy in rotation? Rank the speeds at the bottom and compare them with a block sliding without friction.

Set up: $I = cMR^2$ with $c = 1$ (hoop), $\frac{1}{2}$ (disc), $\frac{2}{5}$ (sphere). Static friction does no work, so $Mgh = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2$ with $\omega = v/R$:

$$Mgh = \tfrac{1}{2}Mv^2(1 + c) \quad \Rightarrow \quad v = \sqrt{\frac{2gh}{1 + c}}.$$

Execute ($h = 0.80\,\mathrm{m}$, $2gh = 15.7\,\mathrm{m^2/s^2}$):

Body	c	v at the bottom
frictionless block (no rotation)	0	$3.96\,\mathrm{m/s}$
solid sphere	0.40	$3.35\,\mathrm{m/s}$
solid disc	0.50	$3.23\,\mathrm{m/s}$
thin hoop	1.00	$2.80\,\mathrm{m/s}$

Evaluate: the ranking is sphere, disc, hoop. The mass M and the radius R cancel, so only the shape (c) matters. Every rolling body is slower than the sliding block because part of the energy goes into rotation. ✓

Question

A light string is wrapped round a solid cylinder of mass M and radius R. You hold the free end fixed and release the cylinder from rest; the string unwinds without slipping. Find (a) the speed of its centre after it has descended a height h (use energy), and (b) its downward acceleration and the tension in the string (use forces and torques).

Two tools, one body

The cylinder both translates and rotates. Energy needs no acceleration; $\sum F = Ma$ and $\sum \tau = I\alpha$ give a and T.

Predict first

Will a be larger or smaller than g? Will the tension be larger or smaller than Mg? Which tool gives the speed faster?

Set up (a): the string is held fixed, so only gravity does work and $v_{\text{cm}} = R\omega$ with $I = \frac{1}{2}MR^2$: $K = \frac{1}{2}Mv^2 + \frac{1}{2}\left(\frac{1}{2}MR^2\right)\dfrac{v^2}{R^2} = \frac{3}{4}Mv^2$.

(a) $Mgh = \frac{3}{4}Mv^2$, so $v = \sqrt{\dfrac{4gh}{3}}$. For $h = 1.0\,\text{m}$: $v = 3.6\,\text{m/s}$.

Set up (b): down positive for the translation, $Mg - T = Ma$; torque of T about the centre, $TR = I\alpha = \frac{1}{2}MR^2\alpha$, with $a = R\alpha$ (no slipping).

(b) $T = \frac{1}{2}Ma$, so $Mg - \frac{1}{2}Ma = Ma$ gives $a = \dfrac{2}{3}g = 6.5\,\text{m/s}^2$ and $T = \dfrac{1}{3}Mg$.

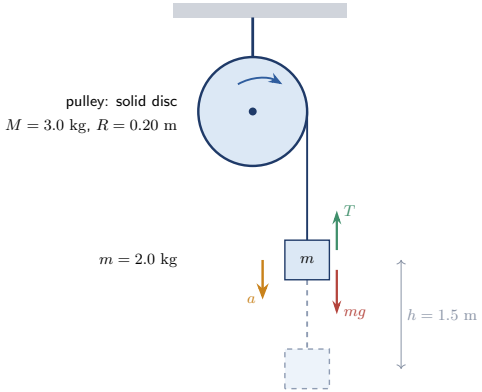
Evaluate: $a < g$ and $T < Mg$ ✓. Check: $v^2 = 2ah = \frac{4}{3}gh$, the same as from energy. ✓The cylinder falls more slowly than a free mass because part of the energy goes into rotation.

Example 10: a mass on a cord turns a pulley

CORE

Question

A $2.0\,\mathrm{kg}$ block hangs from a light cord wrapped round a solid disc pulley ($M = 3.0\,\mathrm{kg}$, $R = 0.20\,\mathrm{m}$) on a frictionless axle. The block is released from rest. Find (a) its acceleration and the tension, and (b) its speed after it has fallen $1.5\,\mathrm{m}$, by energy.



Predict first

Is a larger or smaller than g? Is the tension greater than, equal to, or smaller than $mg = 19.6\,\mathrm{N}$?

Set up: downward positive for the block, the same sense of rotation for the pulley; the cord does not slip, so $a = R\alpha$. $I = \frac{1}{2}MR^2$.

Block: $mg - T = ma$. **Pulley:** $TR = I\alpha = \frac{1}{2}MR^2\frac{a}{R} \Rightarrow T = \frac{1}{2}Ma$.

Combine: $mg = \left(m + \frac{1}{2}M\right)a \Rightarrow a = \dfrac{mg}{m + M/2} = \dfrac{(2.0)(9.80)}{2.0 + 1.5} = 5.60\,\mathrm{m/s^2}$.

$T = \frac{1}{2}Ma = \frac{1}{2}(3.0)(5.60) = 8.40\,\mathrm{N}$.

Evaluate: $a < g$ and $T = 8.4\,\mathrm{N} \ll mg = 19.6\,\mathrm{N}$: the pulley must be accelerated, so the tension is much less than the weight; if T were mg, the block would not accelerate. After falling $1.5\,\mathrm{m}$: $v = \sqrt{2ah} = \sqrt{2(5.60)(1.5)} = 4.10\,\mathrm{m/s}$ and $\omega = v/R = 20.5\,\mathrm{rad/s}$.

Set up: system: block, pulley and Earth. The cord and the axle do no net work on the system, and $\omega = v/R$. State 1: at rest; state 2: after falling $h = 1.5\,\mathrm{m}$ ($y = 0$ at state 2).

Execute: $mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{1}{2}MR^2\right)\frac{v^2}{R^2} = \frac{1}{2}\left(m + \frac{1}{2}M\right)v^2$.

$$v = \sqrt{\frac{2mgh}{m + M/2}} = \sqrt{\frac{2(2.0)(9.80)(1.5)}{3.5}} = \sqrt{16.8} = 4.10\,\mathrm{m/s}.$$

Evaluate: the same $4.10\,\mathrm{m/s}$ as the force method, and smaller than the free-fall value $\sqrt{2gh} = 5.42\,\mathrm{m/s}$ because part of the energy goes into rotating the pulley: $\frac{1}{2}I\omega^2 = \frac{1}{2}(0.060)(20.5)^2 = 12.6\,\mathrm{J}$ of $29.4\,\mathrm{J}$. ✓

Physical meaning

Energy gave v in one line; $\sum\tau = I\alpha$ was needed for a and T. Choose by what is asked.

For a particle with linear momentum $\vec{\boldsymbol{p}} = m\vec{\boldsymbol{v}}$ (the product of mass and velocity) at position $\vec{\boldsymbol{r}}$ from a chosen point O:

Rule: angular momentum

$$\vec{\boldsymbol{L}} = \vec{\boldsymbol{r}} \times \vec{\boldsymbol{p}} \qquad L = mvr\sin\varphi$$

For a rigid body turning about a fixed symmetry axis: $L = I\omega, \quad [L] = \mathrm{kg\,m^2/s}.$

The direction of $\vec{\boldsymbol{L}}$ is given by the right-hand rule, along the axis for rotation about a symmetry axis. If the axis is not a symmetry axis, $\vec{\boldsymbol{L}}$ need not lie along it; the axis component is $L_z = I\omega_z$.

Rule: the rotational form of Newton's second law

$$\sum \vec{\boldsymbol{\tau}}_{\text{ext}} = \frac{\mathrm{d}\vec{\boldsymbol{L}}}{\mathrm{d}t}$$

Physical meaning

The net external torque is the rate of change of angular momentum, as force is the rate of change of momentum. For fixed I: $\mathrm{d}(I\omega)/\mathrm{d}t = I\alpha$, the law of the previous contact.

Question

A $0.50\,\mathrm{kg}$ ball moves in a straight line at $4.0\,\mathrm{m/s}$, passing at a perpendicular distance of $1.2\,\mathrm{m}$ from a point O. (a) Find the magnitude of its angular momentum about O. (b) Does it change as the ball moves along its line? (c) What is L about a point on the ball's line?

Predict first

Which quantity in $mvr\sin\varphi$ changes as the ball moves along the line, and which stay fixed? Is there a torque about O?

(a) $L = mvr\sin\varphi = mvl = (0.50)(4.0)(1.2) = 2.4\,\mathrm{kg\,m^2/s}$, with $l = r\sin\varphi$ the perpendicular distance. (b) No: l and v stay fixed (no force, so no torque about O). (c) $L = 0$, since $l = 0$. **Angular momentum depends on the point chosen.**

Rule: conservation of angular momentum

If the net external torque about an axis is zero, the angular momentum about that axis is constant:

$$\sum \tau_{\text{ext}} = 0 \;\Rightarrow\; L_1 = L_2 \quad \text{so} \quad I_1\omega_1 = I_2\omega_2$$

Physical meaning

Internal forces cancel in pairs, so only external torques change L. The moment of inertia may change during the motion: if I falls, ω must rise.

Conditions

Gravity at the axis, a normal force through the axis and a hinge force at the axis give no torque about that axis. Conservation of L does *not* imply conservation of K.

Answer, with a reason

A skater spinning with her arms out pulls them in. (a) What happens to her angular velocity? (b) To her angular momentum? (c) To her kinetic energy? (d) Where does any extra energy come from?

(a) ω increases, because I decreases. (b) L is unchanged: the ice and gravity exert no net torque about her axis. (c) $K = \frac{1}{2} I\omega^2 = \frac{L^2}{2I}$ *increases* when I decreases. (d) From the chemical energy of her muscles: she does positive work pulling her arms inward.

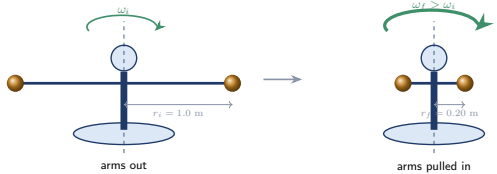
Common error

"The skater turns faster, so energy is conserved." Neither K nor the energy of the arms stays the same; L is the conserved quantity.

Example 11: a person on a rotating stool

Question

A person on a freely turning stool holds a $5.0\,\mathrm{kg}$ mass in each hand and rotates once every $2.0\,\mathrm{s}$ with the arms out ($1.0\,\mathrm{m}$ from the axis). The person and stool alone have $I = 3.0\,\mathrm{kg\,m^2}$ with the arms out and $2.2\,\mathrm{kg\,m^2}$ with the arms in. The arms are pulled in until the masses are $0.20\,\mathrm{m}$ from the axis. Find (a) the new rate of rotation, and (b) the kinetic energy before and after.



Predict first

Will ω at least double? By what factor does the kinetic energy change? Where does the extra energy come from?

A person on a rotating stool: solution

Set up: no external torque about the vertical axis, so $I_i\omega_i = I_f\omega_f$. Treat the masses as point masses.

$I_i = 3.0 + 2(5.0)(1.0)^2 = 13\,\text{kg}\,\text{m}^2$; $I_f = 2.2 + 2(5.0)(0.20)^2 = 2.6\,\text{kg}\,\text{m}^2$; $\omega_i = \dfrac{1\text{ rev}}{2.0\,\text{s}} = 0.50$ rev/s $= 3.14\,\text{rad/s}$.

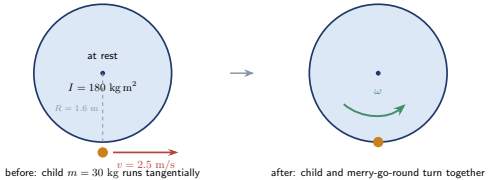
(a) $\omega_f = \omega_i \dfrac{I_i}{I_f} = (0.50\text{ rev/s})\dfrac{13}{2.6} = 2.5$ rev/s $= 15.7\,\text{rad/s}$ (the ratio is the same in rev/s or rad/s).

(b) $K_i = \frac{1}{2}(13)(3.14)^2 = 64\,\text{J}$; $K_f = \frac{1}{2}(2.6)(15.7)^2 = 321\,\text{J}$.

Evaluate: ω rises by a factor $13/2.6 = 5.0$, and K rises by the same factor ($K = \frac{1}{2}L\omega$ with L fixed). The extra $257\,\text{J}$ is the work the person does pulling the masses inward; it is not conserved. ✓

Independent problem

A $30\,\mathrm{kg}$ child runs at $2.5\,\mathrm{m/s}$ along a tangent to a stationary merry-go-round ($I = 180\,\mathrm{kg\,m^2}$, $R = 1.6\,\mathrm{m}$) and jumps on at the rim. Find (a) the angular momentum of the child about the axis before the jump, (b) the angular velocity afterwards, and (c) the fraction of the kinetic energy that is lost.



before: child $m = 30$ kg runs tangentially — after: child and merry-go-round turn together

Predict first

Is ω larger or smaller than $v/R = 1.56\,\mathrm{rad/s}$? Is energy conserved in the jump?

Set up: the axle exerts no torque about the axis, and the jump is brief; L is conserved about the axis. The child is a particle with $l = R$ (tangential motion).

(a) $L = mvR = (30)(2.5)(1.6) = 120\,\mathrm{kg\,m^2/s}$.

(b) $I_{\text{tot}} = I + mR^2 = 180 + (30)(1.6)^2 = 257\,\mathrm{kg\,m^2}$, so $\omega = \dfrac{L}{I_{\text{tot}}} = \dfrac{120}{257} = 0.467\,\mathrm{rad/s}$.

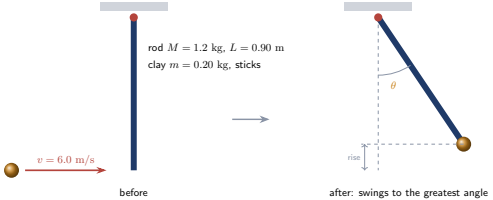
(c) $K_i = \frac{1}{2}(30)(2.5)^2 = 93.8\,\mathrm{J}$; $K_f = \frac{1}{2}(257)(0.467)^2 = 28.0\,\mathrm{J}$; fraction lost $= \dfrac{93.8 - 28.0}{93.8} = 0.70$.

Evaluate: $\omega \ll v/R$, as predicted, because the child shares L with a large I. 70% of the kinetic energy becomes thermal energy (the child's feet and the platform): a sticking collision does not conserve K. ✓

Decide what physics applies; no formula is given

A $0.20\,\mathrm{kg}$ lump of clay moving horizontally at $6.0\,\mathrm{m/s}$ hits the lower end of a uniform rod ($M = 1.2\,\mathrm{kg}$, $L = 0.90\,\mathrm{m}$) hanging from a frictionless hinge at its top, and sticks.

(a) Find the angular velocity just after the impact.
(b) Find the greatest angle of swing.



Predict first

Which quantity is conserved in the impact, which in the swing? Is energy conserved in the impact?

Identify: two stages. *Impact:* short, with a large force from the hinge, but the hinge force has no torque about the hinge, and gravity has none either (it acts along the line through the hinge at that instant): L about the hinge is conserved. *Swing:* only gravity does work: energy is conserved.

(a) $L_i = mvL = (0.20)(6.0)(0.90) = 1.08\,\mathrm{kg\,m^2/s}$.

$I = \frac{1}{3}ML^2 + mL^2 = 0.324 + 0.162 = 0.486\,\mathrm{kg\,m^2}$, so $\omega = \dfrac{1.08}{0.486} = 2.22\,\mathrm{rad/s}$.

(b) $K = \frac{1}{2}I\omega^2 = 1.20\,\mathrm{J}$. The centres of mass rise $\frac{L}{2}(1-\cos\theta)$ (rod) and $L(1-\cos\theta)$ (clay): $1.20 = \left(Mg\frac{L}{2} + mgL\right)(1-\cos\theta) = (5.29 + 1.76)(1-\cos\theta) \Rightarrow \cos\theta = 0.830,\ \theta = 33.9^\circ$.

Evaluate: the clay had $3.6\,\mathrm{J}$ before and $1.2\,\mathrm{J}$ after: $2.4\,\mathrm{J}$ became thermal energy in the impact, so energy could not be used there; and L could not be used in the swing, because gravity exerts a net torque about the hinge. ✓

Connect: where rotation is used next

Gravitation, Lecture 6

Kepler's second law (equal areas in equal times) is conservation of angular momentum (supplementary notes).

Lecture 11

Rotational kinetic energy $\frac{1}{2}I\omega^2$ is the share of a diatomic molecule's energy that sets its heat capacity.

Lecture 12

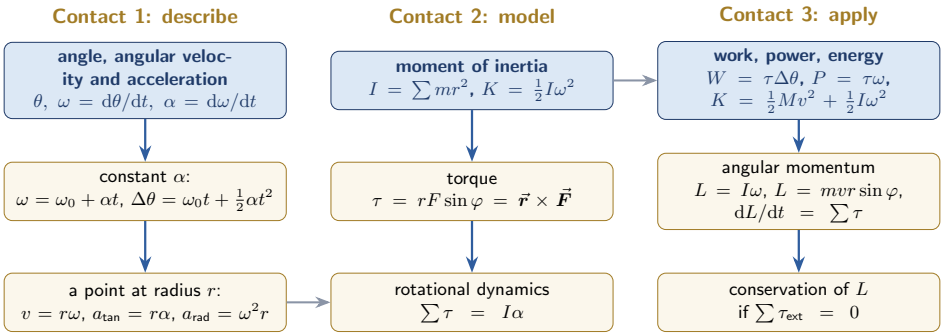
Engine power is torque times angular speed, $P = \tau\omega$; efficiency follows from the second law.

Physical meaning

Rotation reuses every linear idea: $F \to \tau$, $m \to I$, $p \to L$, $W = Fx \to \tau\theta$. The conservation laws of this course now come as a set.

Common mistakes and the correct habit

Mistake	Instead
Using the angle between $\vec{F}$ and the horizontal for torque.	The angle is between $\vec{r}$ and $\vec{F}$, or use the lever arm.
Writing torque in joules.	Torque is in N m; energy is in J.
Assuming zero net force means zero net torque.	A couple turns a body with no net force.
Assuming the tension equals the weight on a pulley with mass.	Write $\sum F = ma$ and $\sum \tau = I\alpha$ with $a = R\alpha$.
Leaving the rotational term out of the energy equation.	$K = \frac{1}{2}Mv_{\text{cm}}^2 + \frac{1}{2}I\omega^2$.
Conserving L when an external torque acts, or assuming K is conserved.	List the external torques; compare K_1 and K_2.
Forgetting that L depends on the point chosen.	Name the point or axis for every $\vec{L}$ and τ.
Saying static friction does work in rolling, or using $v_{\text{cm}} = R\omega$ for a slipping wheel.	The contact point of a rolling wheel is momentarily at rest: no work; $v_{\text{cm}} = R\omega$ holds only without slipping.



each idea has a linear partner: $x, v, a \leftrightarrow \theta, \omega, \alpha$; $m \leftrightarrow I$; $F \leftrightarrow \tau$; $p \leftrightarrow L$

MASTER PROCEDURE – ROTATION

1. IDENTIFY The body, the axis, and what is asked: an angle, ω, α, I, a torque, an energy, a power or an angular momentum.
2. SKETCH The axis, the positive sense, every force with its point of application, the states before and after.
3. MODEL Kinematics, $\sum \tau = I\alpha$, energy and work, or angular momentum; assumptions (rigid body, fixed axis, no slipping).
4. PREDICT Signs, direction, whether ω rises or falls, whether K rises or falls, and the rough size.
5. SELECT THE METHOD $\sum \tau = I\alpha$ for accelerations and forces; energy for speeds between two states; L when no external torque acts.
6. EXECUTE Radians throughout; τ and I about the same axis (the seven toolkits are in the reference deck).
7. EVALUATE Units, sign, magnitude, limiting cases, and a second method.

SKETCH $\rightarrow$ PREDICT $\rightarrow$ CALCULATE $\rightarrow$ CHECK.

Inside the course framework

IDENTIFY $\leftrightarrow$ step 1
SET UP $\leftrightarrow$ steps 2–5
EXECUTE $\leftrightarrow$ step 6
EVALUATE $\leftrightarrow$ step 7

Physical meaning

Next: the Gravitation lecture. Kepler's second law (equal areas in equal times) is conservation of angular momentum for a planet.

Summary: key ideas

Key ideas

- Every point has the same ω and α; $v = r\omega$.
- I depends on the axis and on how the mass is spread.
- $\sum \tau = I\alpha$ is Newton's second law for rotation.
- Rolling bodies have translational and rotational K.
- With no external torque, L is conserved; K need not be.

Where to look

Definitions: the vocabulary slides of each contact; equation anatomy and the seven toolkits are in the reference deck.

Common mistakes: the last content slide of each contact.

Connections: the Connect slide.

Summary: core equations

Idea	Equation	Conditions and units
Angle, ω, α	$\theta = s/r, \quad \omega = \mathrm{d}\theta/\mathrm{d}t, \quad \alpha = \mathrm{d}\omega/\mathrm{d}t$	rad, rad/s, rad/s^2
Constant α	$\omega = \omega_0 + \alpha t, \quad \theta - \theta_0 = \omega_0 t + \frac{1}{2}\alpha t^2$	only if α constant
Point on a body	$v = r\omega, \quad a_{\mathrm{tan}} = r\alpha, \quad a_{\mathrm{rad}} = \omega^2 r$	radians
Inertia	$I = \sum m r^2, \quad I = I_{\mathrm{cm}} + Md^2$	$\mathrm{kg\,m^2}$
Torque	$\tau = rF\sin\varphi, \quad \sum \tau = I\alpha$	same axis; N m
Work, power	$W = \tau\Delta\theta, \quad P = \tau\omega$	J, W
Energy	$K = \frac{1}{2}Mv_{\mathrm{cm}}^2 + \frac{1}{2}I\omega^2$	rolling: $v_{\mathrm{cm}} = R\omega$
Angular momentum	$L = I\omega, \quad \sum \tau_{\mathrm{ext}} = \mathrm{d}L/\mathrm{d}t$	$\mathrm{kg\,m^2/s}$
Conservation	$I_1\omega_1 = I_2\omega_2$	no external torque

Checklist: can I . . . CORE

- ☐ convert between degrees, radians, revolutions and rpm?
- ☐ find ω and α from $\theta(t)$ and use the constant-α equations?
- ☐ relate v, a_{tan} and a_{rad} to ω and α?
- ☐ calculate I for particles, standard bodies and parallel axes?
- ☐ calculate a torque, with its sign, and a net torque?
- ☐ use $\sum \tau = I\alpha$ with a cord, a pulley or a hinge?
- ☐ calculate the work done by a torque, the power and $K = \frac{1}{2} I\omega^2$?
- ☐ use the energy equation with rotation, including rolling?
- ☐ state the condition for conservation of angular momentum?
- ☐ use $L = I\omega$ and $L = mvl$, and explain what happens to K?
- ☐ choose between $\sum \tau = I\alpha$, energy and L in an unfamiliar problem?

After this week

Tutorial sheet: extra problems on every skill, the old-deck and moved problems, and a challenge with no method given. Reference deck: equation anatomy and the seven toolkits. Supplementary notes: radians and units, the proofs of $K = \frac{1}{2} I\omega^2$ and the parallel-axis theorem, the integrals for I, rolling dynamics, and Kepler's second law.

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